Statistics and Trends: Exploring a Diabetes Dataset

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Introduction

Statistics play an important role in healthcare because they help us understand different health measurements and how they are related to patient outcomes. In this report, I have used a diabetes dataset containing information from 442 participants. The main purpose of this analysis is to study how the outcome values are distributed and to explore the relationship between BMI (Body Mass Index) and the outcome measure. I have also compared the results across different BMI groups to identify any noticeable differences. Through this analysis, I aim to understand the patterns within the dataset. The findings are only based on the available sample data and should not be used to make medical diagnoses or treatment decisions.

1. Distribution of the Outcome Measure

After looking at the histogram, I found that the average outcome value is 152.1, while the median is 140.5. The standard deviation is 77.1, and the values range from 25 to 346. The mean and median are quite close to each other, but the graph shows that the values are spread over a wide range. This means that the participants in the dataset have different outcome scores. The histogram makes it easier to see how the values are distributed across the sample. However, it does not tell us the exact reasons for the differences between participants.

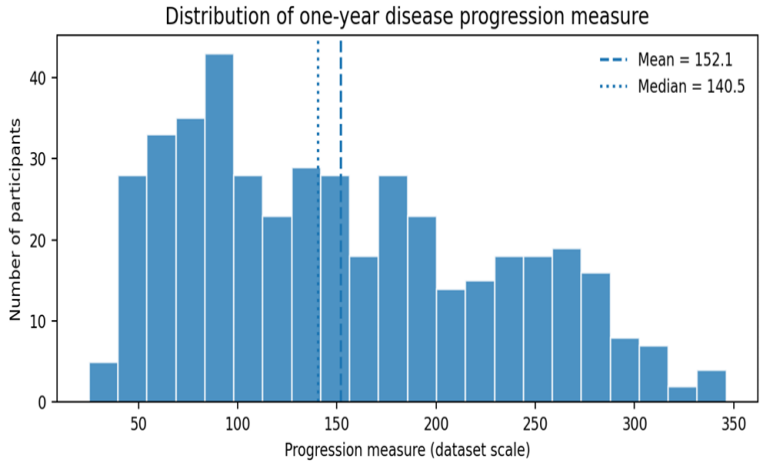


Figure 1. Distribution of the one-year progression measure.

2. BMI and the Outcome Measure

When I looked at the scatter plot, I noticed that BMI and the outcome score have a positive relationship. The Pearson correlation value is 0.59, which shows a moderate positive connection between these two measurements. In simple terms, participants with higher BMI values generally tend to have higher progression scores as well. However, the points on the graph are spread out rather than being close to the trend line. This suggests that BMI may be related to the outcome, but it is not the only factor that could affect the results. Other health-related and lifestyle factors may also play a role. It is important to remember that this relationship does not mean that a higher BMI directly causes an increase in the progression score.

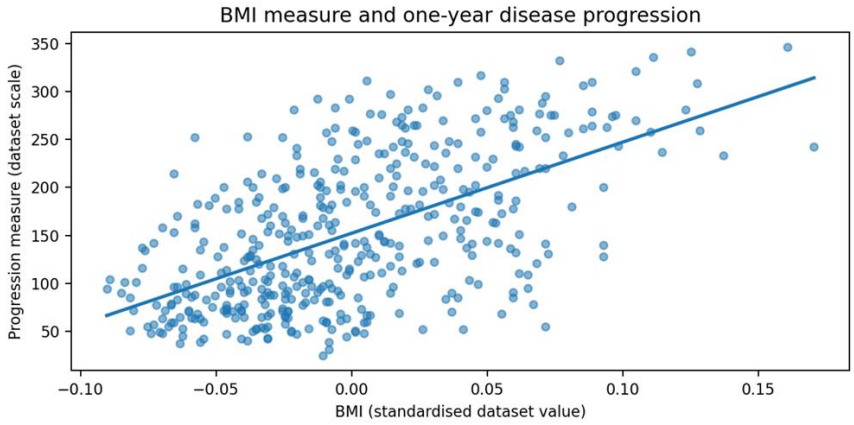


Figure 2. Scatter plot showing BMI and the progression measure.

3. Comparing BMI Groups

To make the results easier to understand, I divided the participants into four groups according to their BMI values. The box plot helps me compare the outcome scores across these groups. It shows the middle value (median), how the scores are spread out and whether there are any unusually high or low values. These groups are created based on the BMI values in this dataset and are not based on official medical BMI categories. From the graph, I noticed that participants in the higher-BMI groups generally have higher progression scores. However, there is some overlap between the groups, which means that the differences are not completely clear. This suggests that BMI and the outcome score may be related, but the pattern is not the same for every participant.

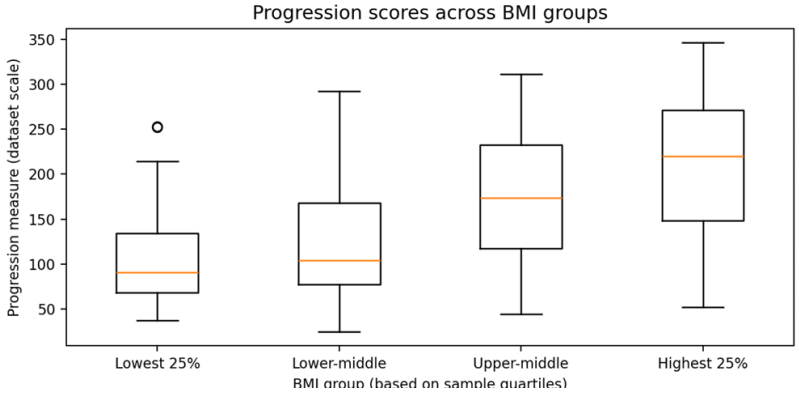


Figure 3. Outcome scores across four sample-based BMI groups

Conclusion

After completing this analysis, I found that the one-year progression scores vary quite a lot between participants. The results also show a moderate positive relationship between BMI and the outcome score. When comparing the four BMI groups, I noticed that participants with higher BMI values generally had higher progression scores, although there were some similarities and overlaps between the groups. The heatmap also showed that some serum measurements have a relationship with the outcome. This helped me understand that health-related results can be influenced by several different factors rather than just one measurement. Overall, this analysis helped me learn how graphs and statistical methods can be used to identify patterns in healthcare data.

Git-Hub Link : https://github.com/mandlavamsee/Diabetes-Statistics-and-Trends